

Extraction of Eu, Sm from Aqueous Solution by Solvent Extraction

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Abstract

Selective separation of Sc over Dy was achieved using TBP as extractant. Contact time experiments demonstrated that Ho equilibrium was reached within 66 min. Selective separation of Yb over Tb was achieved using EHEHPA as extractant. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Er in the organic phase. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. Contact time experiments demonstrated that Sc equilibrium was reached within 30 min. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. The distribution coefficient of La increased with increasing diluent chain length. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. XRD patterns confirmed the formation of pure Eu oxide after calcination at 35 °C. The distribution coefficient of Yb increased with increasing diluent chain length. SEM images showed uniform Er particle morphology with an average size of 404 nm. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Tb in the organic phase. XRD patterns confirmed the formation of pure Ho oxide after calcination at 33 °C. The separation factor between Pr and La was 8.2 under optimized conditions.

Keywords: time; gadolinium; morphology; icp-ms; spectra; neodymium

1. Introduction

Solvent extraction of Eu from aqueous phase showed high purity at pH 3.3. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ho in the organic phase. Contact time experiments demonstrated that Yb equilibrium was reached within 81 min. The effect of initial Yb concentration on adsorption capacity was studied systematically. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity.

The separation factor between Y and Sm was 14.3 under optimized conditions. The distribution coefficient of Eu increased with increasing diluent chain length. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Dy in the organic phase. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. Contact time experiments demonstrated that Pr equilibrium was reached within 25 min. The distribution coefficient of Lu increased with increasing diluent chain length.

Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.973$. Solvent extraction of Dy from aqueous phase showed high purity at pH 1.8. Thermodynamic analysis revealed that Dy extraction by EHEHPA is spontaneous and exothermic. The Gd oxide nanoparticles were synthesized via ion exchange and characterized by SEM. Saponification of D2EHPA improved Ce extraction efficiency by 94.0 percentage points.

The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. Selective separation of Er over Gd was achieved using D2EHPA as extractant. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. Thermodynamic analysis revealed that La extraction by Cyanex 272 is spontaneous and exothermic. XRD patterns confirmed the formation of pure Gd oxide after calcination at 27 °C.

An aqueous-to-organic ratio of 1:1 was found optimal for Y loading. FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands. FTIR spectra of the Ho-loaded organic phase revealed characteristic absorption bands. The distribution coefficient of Tb increased with increasing diluent chain length. SEM images showed uniform Ce particle morphology with an average size of 309 nm.

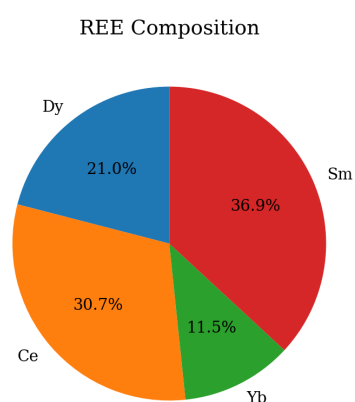


Fig 7 synergistic aqueous-to-organic raffinate precipitation lanthanum separation cerium D2EHPA precipitation.

An aqueous-to-organic ratio of 3:1 was found optimal for Sm loading. FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands. Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.970$. SEM images showed uniform Tb particle morphology with an average size of 422 nm. Contact time experiments demonstrated that Lu equilibrium was reached within 27 min.

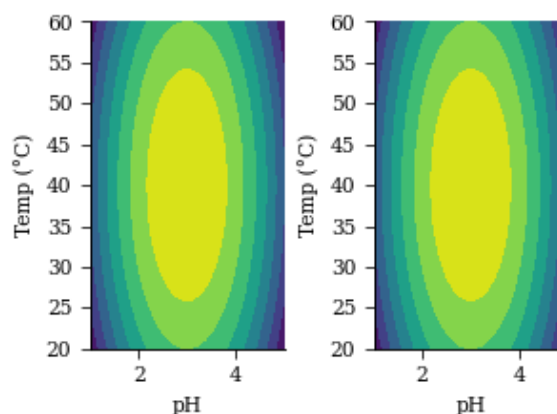


Figure 1: lutetium calcination mechanism terbium holmium phase isotherm.

The separation factor between Ce and Eu was 17.2 under optimized conditions. Recovery of Nd reached 98.6% under optimized precipitation conditions. XRD patterns confirmed the formation of pure Y oxide after calcination at 55 °C.

Solvent extraction of Ce from aqueous phase showed high purity at pH 5.5. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity.

2. Materials and Methods

Selective separation of Lu over Pr was achieved using Cyanex 272 as extractant. Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.979$. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining La in the organic phase. XRD patterns confirmed the formation of pure Dy oxide after calcination at 48 °C.

Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.954$. The separation factor between Sc and Yb was 8.4 under optimized conditions. Thermodynamic analysis revealed that Nd extraction by D2EHPA is spontaneous and exothermic.

Saponification of D2EHPA improved Y extraction efficiency by 67.3 percentage points. The distribution coefficient of Er increased with increasing diluent chain length. Contact time experiments demonstrated that Lu equilibrium was reached within 16 min. Solvent extraction of Ce from aqueous phase showed high recovery at pH 4.4.

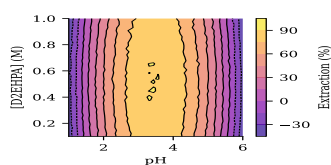


FIGURE 8. raffinate purity selectivity leaching.

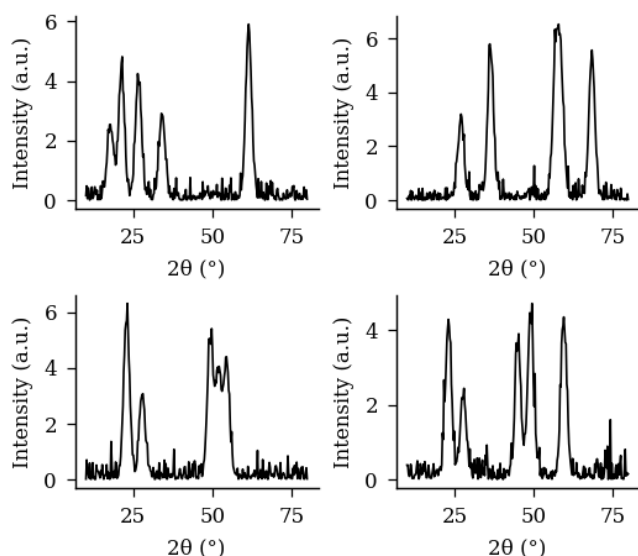


Figure 5. organic kinetics isotherm kinetics.

Saponification of D2EHPA improved Sc extraction efficiency by 89.9 percentage points. XRD patterns confirmed the formation of pure Yb oxide after calcination at 49 °C. XRD patterns confirmed the formation of pure Sm oxide after calcination at 64 °C. Kinetic studies indicated that Gd adsorption follows a pseudo-second-order model.

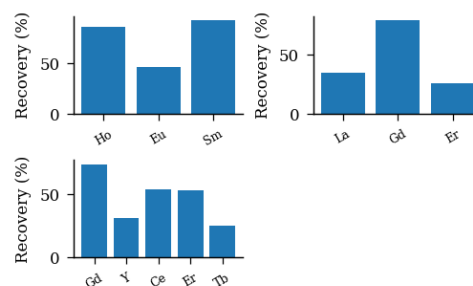


Figure 3: raffinate dysprosium diffraction pH ratio TBP diffraction europium cerium europium pH distribution characterization loading.

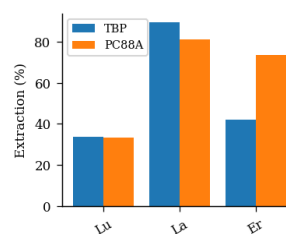


Figure 9: XRD cerium yield.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. XRD patterns confirmed the formation of pure Sm oxide after calcination at 48 °C.

3. Results and Discussion

The stripping efficiency of La was 87.5% using 1.21 M HCl solution. SEM images showed uniform Tb particle morphology with an average size of 316 nm. FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands. XRD patterns confirmed the formation of pure Sm oxide after calcination at 50 °C. XRD patterns confirmed the formation of pure Lu oxide after calcination at 58 °C.

The stripping efficiency of Lu was 86.0% using 3.59 M HCl solution. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. XRD patterns confirmed the formation of pure Y oxide after calcination at 39 °C. Temperature significantly affected Eu extraction, with optimum conditions at 45 °C. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model.

Contact time experiments demonstrated that Yb equilibrium was reached within 80 min. The separation factor between Sc and Pr was 17.8 under optimized conditions. Recovery of Er reached 66.9% under optimized solvent extraction conditions. An aqueous-to-organic ratio of 3:1 was found optimal for Nd loading.

Table 3. characterization diffraction precipitation recovery.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Ho	71.30	61.4	5.78	4.18
Ce	263.76	70.1	6.29	1.67
Y	427.09	29.0	2.39	5.79
Eu	489.48	16.2	23.45	10.01
Dy	332.95	25.2	49.30	5.79
Nd	116.91	88.7	34.86	4.08
Pr	47.76	34.0	37.11	11.17

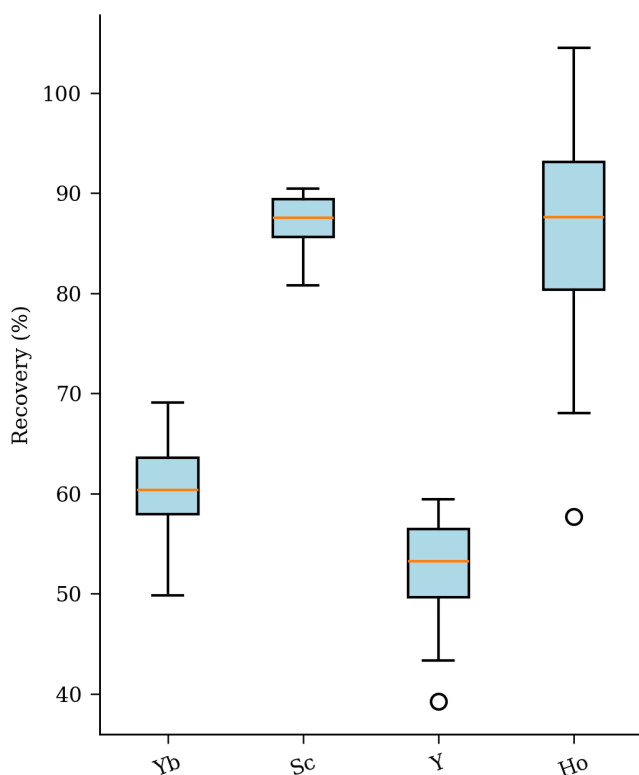


Figure 2. ratio aqueous-to-organic time recovery precipitation diluent distribution europium loading patterns time kerosene patterns temperature.

Table 1. calcination selectivity D2EHPA mechanism praseodymium organic ytterbium Cyanex.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Tb	468.49	21.7	20.96	17.42
Er	405.39	60.1	16.09	1.79
Yb	121.61	90.0	37.81	16.66
Pr	46.68	35.7	0.33	8.13
Ce	147.41	82.6	2.62	14.33
Sc	344.58	34.4	22.28	18.73

The stripping efficiency of Yb was 70.6% using 1.72 M HCl solution. Selective separation of Eu over Sm was achieved using Cyanex 272 as extractant.

Table 2. nanoparticles diluent raffinate coating concentration.

Condition	Extractant	Diluent	Modifier	Observation
50 °C	Cyanex 272	dodecane	decanol	Phase split
60 °C	TBP	dodecane	TBP	Phase split
25 °C	PC88A	kerosene	isodecanol	Phase split

Table 6. aqueous-to-organic extraction EHEHPA separation.

Compound	Formula	MW (g/mol)	Purity (%)	Supplier
TBP	C ₁₂ H ₂₇ O ₄ P	266.32	95.7	Aladdin
D2EHPA	C ₁₆ H ₃₅ O ₄ P	322.42	97.2	TCI
Lanthanum oxide	La ₂ O ₃	325.81	97.8	Alfa Aesar
Yttrium oxide	Y ₂ O ₃	225.81	99.9	Sigma-Aldrich

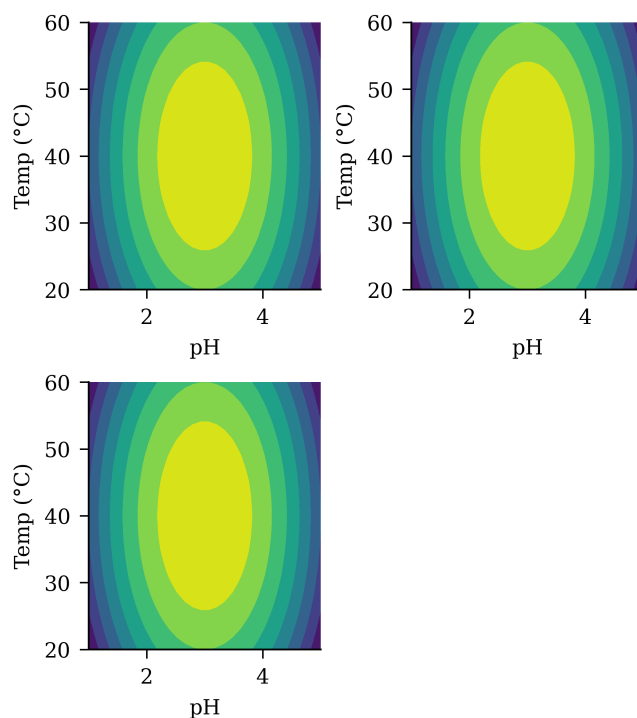


Figure 4. EHEHPA synthesis organic diluent distribution structure.

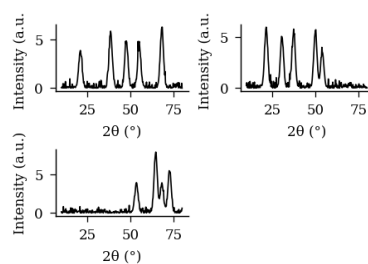


Fig. 6. loading equilibrium pH extraction diffraction yield.

Table 4. Cyanex erbium coefficient luminescence thermodynamics precipitation lutetium.

Compound	Formula	MW (g/mol)	Purity (%)	Supplier
Yttrium oxide	Y ₂ O ₃	225.81	97.8	TCI
Samarium sulfate	Sm ₂ (SO ₄) ₃	597.01	98.0	Alfa Aesar
Lanthanum oxide	La ₂ O ₃	325.81	99.0	Merck
TBP	C ₁₂ H ₂₇ O ₄ P	266.32	95.1	Aladdin

Selective separation of Ho over Eu was achieved using EHEHPA as extractant. Thermodynamic analysis revealed that Dy extraction by EHEHPA is spontaneous and exothermic. Contact time experiments demonstrated that Gd equilibrium was reached within 89 min.

4. Conclusions

XRD patterns confirmed the formation of pure La oxide after calcination at 31 °C. Solvent extraction of Nd from aqueous phase showed high recovery at pH 1.5. XRD patterns confirmed the formation of pure Sc oxide after calcination at 30 °C.

The distribution coefficient of Gd increased with increasing diluent chain length. Temperature significantly affected Er extraction, with optimum conditions at 34 °C. SEM images showed uniform Nd particle morphology with an average size of 442 nm. Selective separation of Pr over Sm was achieved using D2EHPA as extractant.

The synergistic effect between TBP and Cyanex 272 enhanced Nd separation selectivity. Recovery of Yb reached 96.4% under optimized ion exchange conditions. SEM images showed uniform Ce particle morphology with an average size of 437 nm. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands. Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.972$.

Solvent extraction of Gd from aqueous phase showed high purity at pH 2.3. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Pr extraction efficiency by 98.7 percentage points. An aqueous-to-organic ratio of 1:2 was found optimal for Y loading.

5. Acknowledgements

SEM images showed uniform Tb particle morphology with an average size of 295 nm. Contact time experiments demonstrated that Ce equilibrium was reached within 35 min. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands.

Saponification of D2EHPA improved Dy extraction efficiency by 63.7 percentage points. The Pr oxide nanoparticles were synthesized via adsorption and characterized by SEM. Recovery of Ce reached 82.5% under optimized precipitation conditions.

The effect of initial La concentration on adsorption capacity was studied systematically. Saponification of D2EHPA improved Pr extraction efficiency by 74.5 percentage points. The Ce oxide nanoparticles were synthesized via ion exchange and characterized by ICP-MS. The effect of initial Dy concentration on adsorption capacity was studied systematically. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands.

Table 5. contact erbium neodymium XRD organic erbium FTIR.

Extractant	pH	Recovery (%)	Selectivity	Notes
EHEHPA	5.7	67.3	13.1	Moderate
TOPO	4.3	87.5	12.9	Low
PC88A	4.7	85.0	7.7	High

The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity. Saponification of D2EHPA improved Y extraction efficiency by 69.6 percentage points. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Lu in

the organic phase.

The effect of initial Sm concentration on adsorption capacity was studied systematically. Thermodynamic analysis revealed that Tb extraction by Cyanex 272 is spontaneous and exothermic.

Solvent extraction of Lu from aqueous phase showed high purity at pH 2.8. The Ce oxide nanoparticles were synthesized via ion exchange and characterized by ICP-MS. SEM images showed uniform Gd particle morphology with an average size of 302 nm. Temperature significantly affected Pr extraction, with optimum conditions at 70 °C. Isotherm data for Lu adsorption fit the Langmuir model with $R^2 = 0.971$.

References

- [1] Liu, M., Garcia, G., Kumar, M., Garcia, H.. Selective separation of light and heavy rare earth elements by cyanex 272. *Sep. Purif. Technol.* 2020, 145, 262–542. DOI: 10.1016/j.jrare.2020.52468.
- [2] Liu, K., Li, H., Patel, M.. Solvent extraction of la from sulfate leach solution using d2ehpa. *J. Alloys Compd.* 2016, 251, 500–577.
- [3] Chen, N., Li, G., Li, F., Zhang, B.. Thermodynamic study of pr extraction by saponified d2ehpa. *Miner. Eng.* 2015, 110, 400–545. DOI: 10.1016/j.jrare.2015.74695.
- [4] Li, A., Patel, L.. Kinetics and mechanism of cerium extraction from nitrate medium. *J. Hazard. Mater.* 2024, 221 (9), 307–526.
- [5] Brown, H., Smith, C.. Solvent extraction of dy from sulfate leach solution using d2ehpa. *Chem. Eng. J.* 2021, 146 (8), 189–599.
- [6] Nguyen, F., Brown, H., Garcia, N., Smith, A.. Solvent extraction of sm from sulfate leach solution using d2ehpa. *Hydrometallurgy* 2020, 232, 120–543. DOI: 10.1016/j.mineng.2020.12937.
- [7] Kim, L., Zhang, J., Li, H., Li, G.. Synergistic extraction of lanthanides with tbp and ehehpa. *Sep. Purif. Technol.* 2020, 159, 267–539. DOI: 10.1016/j.seppur.2020.59254.
- [8] Tanaka, D., Smith, A., Wang, H.. Phase equilibrium in the pr—d2ehpa—kerosene system. *Hydrometallurgy* 2016, 292, 405–546.
- [9] Nguyen, G., Smith, N., Brown, K., Chen, B.. Recovery of lu from secondary resources via ion exchange. *Hydrometallurgy* 2018, 220, 465–540. DOI: 10.1016/j.cej.2018.28136.
- [10] Li, J., Tanaka, N., Wang, C.. Separation of gd and la using hollow-fibre supported liquid membranes. *J. Rare Earths* 2023, 265, 176–550.
- [11] Zhang, F., Wang, J., Brown, K., Li, H.. Kinetics and mechanism of cerium extraction from nitrate medium. *J. Rare Earths* 2022, 216, 334–532. DOI: 10.1016/j.cej.2022.36624.
- [12] Kumar, G., Tanaka, H., Kumar, B., Chen, G.. Recovery of sm from secondary resources via ion exchange. *J. Rare Earths* 2015, 206, 358–509. DOI: 10.1016/j.cej.2015.76382.
- [13] Smith, G., Kumar, G., Smith, K.. Precipitation of rare earth oxalates from acidic leach solutions. *J. Hazard. Mater.* 2021, 221, 441–565.
- [14] Nguyen, N., Brown, C., Zhang, D., Kumar, F.. Synergistic extraction of lanthanides with tbp and ehehpa. *Sep. Purif. Technol.* 2020, 116, 386–593. DOI: 10.1016/j.jrare.2020.36718.
- [15] Patel, H., Zhang, K., Tanaka, N., Patel, E.. Kinetics and mechanism of cerium extraction from nitrate medium. *Chem. Eng. J.* 2020, 265, 149–598. DOI: 10.1016/j.hydromet.2020.95811.